

Original Research



Land conversion in the tropics from primary forest to agriculture has altered soil hydrologic processes. We estimated saturated hydraulic conductivity (K_s) across a dry, tropical, riparian vegetation gradient in Nicaragua, taking into account soil properties and livestock impact. We found that leaf area index (LAI) had the greatest correlation to K_s , followed by hoofprint density (0.291) and clay content (0.291).

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Woody Vegetation Increases Saturated Hydraulic Conductivity in Dry Tropical Nicaragua

R.J. Niemeyer,* A.K. Fremier, R. Heinse, W. Chávez, and F.A.J. DeClerck

Land conversion in the tropics from primary forest to agricultural land has altered soil hydrologic processes. Woody vegetation is known to increase infiltration rates and saturated hydraulic conductivity (K_s) in primary forests compared with agricultural land, but it is less clear if this relationship holds for a gradient of woody vegetation. In addition, the mechanisms for the effect of woody vegetation on K_s have yet to be fully examined. To quantify the effect of woody vegetation structure on vadose zone hydrology, we estimated K_s in 15 plots across a dry tropical riparian vegetation gradient in Nicaragua, taking into account covariates such as soil properties and livestock impact. Using single linear regression, we found that leaf area index (LAI) had the greatest correlation coefficient of 0.331 to K_s , followed by hoofprint density (0.291) and clay content (0.291). Furthermore, the relationship between LAI and K_s was greater for finer soils than for coarser soils. We found that a forest soil had eight times more preferential flow paths than a pasture soil, and most of these were root-initiated flow paths, suggesting a possible mechanism for the positive correlation between LAI and K_s . We show that the K_s predictions with a pedotransfer function could be improved by incorporating LAI. Our findings support the importance of preserving woody vegetation in key areas on the landscape to maintain hydrologic functions of tropical soils and ecosystems.

Abbreviations: AIC, Akaike information criterion; BA, basal area; BD, bulk density; LAI, leaf area index; PTF, pedotransfer function; SOM, soil organic matter.

Widespread forest conversion to agriculture in the tropics alters vadose zone hydrology. Forest removal and land conversion has resulted in reduced water infiltration (Mapa, 1995; Bruijnzeel, 2000) and saturated hydraulic conductivity (K_s) (Giertz and Dieckkrüger, 2003; de Moraes et al., 2006; Mehta et al., 2008), which can lead to significant increases in surface runoff, erosion, peak flow in rivers, and decreased dry-season base flows (Sandström, 1995; Bruijnzeel, 1989, 2004; Chaves et al., 2008), with far-reaching impacts on human populations.

In searching for the causes of these changes in the hydrology of the vadose zone, surface soil compaction and increased bulk density following tree removal (Alegre and Cassel, 1996; Schack-Kirchner et al., 2007) and intensive grazing (Mehta et al., 2008) have been shown to reduce infiltration rates and K_s . While these studies are informative, they often compare primary forest and agriculture. Instead of all forests being primary forest, secondary forest is emerging as one of the dominant land cover types in the tropics (Giambelluca, 2002; Drigo, 2004; Cuo et al., 2008), and studies have shown that vegetation regrowth following pasture has significantly higher infiltration rates and K_s than the previous pasture (Lal, 1996; Zimmermann and Elsenbeer, 2008). In addition, areas in the tropics where agriculture occurs are complex and can include intensely managed agricultural lands as well as seminatural systems such as savannah-like silvopastoral and multistrata agroforestry systems that contain native trees amidst pasture or crops (Harvey et al., 2006).

To better understand vadose zone hydrology changes in complex tropical landscapes, several studies to date have sought to establish a relationship between infiltration and K_s

across a gradient of deforestation and land use. Giambelluca (2002) showed that in deforested patches, patches that had been deforested at an earlier date, and thereby had more secondary vegetation, had higher K_S than more recently deforested patches with less secondary vegetation. In an Amazonian basin, higher K_S values were found in primary forests, lower K_S in banana (*Musa × paradisiaca* L.), secondary forest, and teak (*Tectona grandis* L. f.) plantations, and lowest K_S in pastures (Zimmermann et al., 2006). In a complex landscape in the Western Ghats of India, in two of the three dominate soils, K_S changed across woody vegetation classes, with the highest in secondary forest, to degraded forest, to plantations (Bonell et al., 2010).

Despite these informative studies, understanding these changes in infiltration and K_S in tropical landscapes with complex soil and vegetation patterns remains elusive. Bonell et al. (2010) saw K_S decreasing with less woody vegetation in Alfisols and Ultisols but not in Vertisols. While Bonell et al. (2010) found that an increasing relationship between K_S and vegetation was specific to soil type, Hassler et al. (2011) and Sobieraj et al. (2002) found that soil type did not principally determine K_S . They found that K_S did not vary across soil types with a range of clay contents in tropical rainforests of Panama and Brazil, respectively. Thompson et al. (2010) used both observational data from several North Carolina field sites and data from a meta-analysis of studies around the world to determine if biomass increased infiltration capacity. They concluded that biomass exerted a primary influence on the infiltration capacity for xeric sites, whereas soil type exerted a primary influence on the infiltration capacity in mesic and hydric sites. These studies point to the importance of soil type and woody vegetation in determining K_S , but clearly the effects and interactions between these factors are still not well understood.

In addition to a need to better understand the strength and complex interactions between soil type and woody vegetation, the mechanisms by which these factors affect K_S are not well established. Several studies in the tropics have cited bioturbation as a driving factor in soil morphological and physical properties (McDonough et al., 2000; Nkem et al., 2000). Increased macroporosity from bioturbation due to roots and faunal activity have been cited as possible mechanisms that cause high K_S in forests in the tropics (Sobieraj et al., 2002; Hassler et al., 2011), yet these mechanisms are rarely tested in tropical field studies. Similarly, woody vegetation is often not incorporated in K_S predictions, hydrologic models, or other predictive tools. For example, pedo-transfer functions (PTFs) such as the Rosetta software (Schaap et al., 2001) are used to predict soil hydraulic properties (e.g., K_S) with soil physical data but often do not include vegetation metrics to make predictions. In the past, Wösten et al. (2001) suggested that other factors that control soil structure, such as biotic activity, could be used to better predict K_S with PTFs. Some studies in temperate regions have shown that vegetation can improve the ability of PTFs to predict K_S (Sharma et al.,

2006; Jana and Mohanty, 2011). If woody vegetation is an important factor in determining K_S , it would be important to include woody vegetation in a PTF.

A better understanding of the drivers and mechanisms for differences in K_S in complex tropical landscapes is crucial to advance watershed management and the restoration of soil hydrologic functions in tropical regions. In particular, there is a need to further elucidate the complex interactions between woody vegetation, soil properties, and land use on K_S in tropical areas and especially in the dry tropics, which lack hydrologic studies compared with the humid tropics (Mbagwu, 1997). To fill this gap, we aimed to: (i) determine the effect of woody vegetation, soil properties and livestock on K_S and elucidate the interactions among these factors; (ii) determine if leaf area index (LAI) could be used to improve the prediction of K_S with a PTF; and (iii) explore root and faunal turbation as a possible mechanism for woody vegetation increasing K_S .

Materials and Methods

Study Site

This study was conducted within the dry tropical watershed of Gil Gonzalez River in the department of Rivas, Nicaragua (Fig. 1). The watershed is 6700 ha and flows into Lake Nicaragua, which ultimately empties into the Caribbean Sea. The watershed is a mixture of pasture, crops, and secondary forest of various tree densities. Using remotely sensed data, a classification of land uses by Chávez (2011) in a portion of the watershed showed that open pastures occupy 45% of the total area, followed by silvopastoral systems with trees (31%), forests (12%), crops (5%), and the remaining 7% as municipal areas and other land uses. Agricultural crops include plantain (*Musa* spp.), rice (*Oryza sativa* L.), and sugarcane (*Saccharum officinarum* L.). Primary and secondary forests are most commonly found on steeply sloped areas and riparian forests along active river channels (D. Sainchez, personal communication, 2010).

Soils in the study area are a part of the Rivas complex derived from Tertiary-age marine parent material composed of clays and sands of varying thickness. Vertisols, which swell during the wet season and shrink to form deep cracks during the dry season, dominate the landscape (D. Sainchez, personal communication, 2010). The climate of the region is characterized by a pronounced wet season from May to November and a dry season from December to April, with the majority of the 1300 mm of precipitation occurring during the wet season (World Meteorological Organization, 1996). Pan evaporation in a previous study at Victoria de Julio, 70 km from Rivas, was 2586 mm/yr averaged across 9 yr (van den Broek et al., 2001), which is probably comparable to pan evaporation at Rivas.

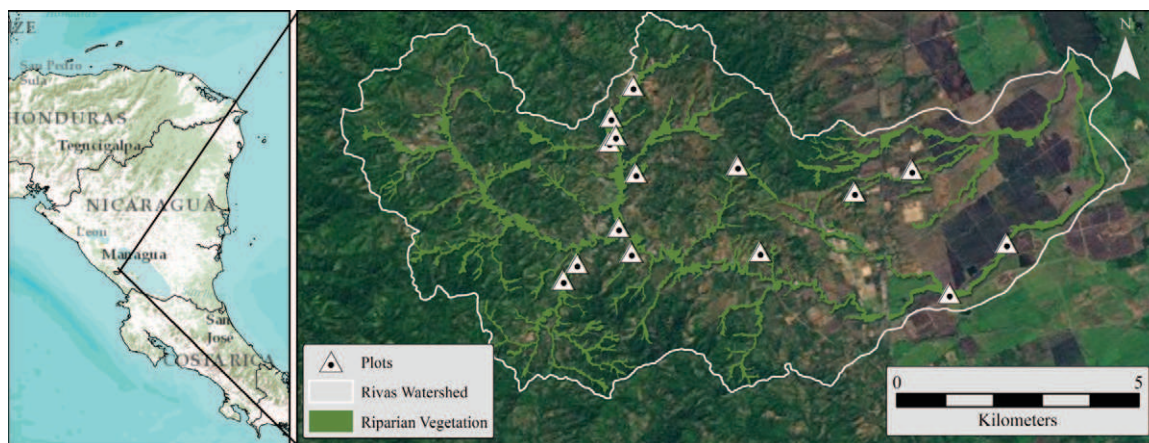


Fig. 1. Map of Nicaragua (left) and watershed of Gil Gonzalez River in the department of Rivas, Nicaragua (right), with plot locations (triangles) and riparian vegetation (green area)

Plot Selection

We selected 15 plots to cover a range of woody vegetation densities, soil textures, and livestock use intensity. To find plots with a range in woody vegetation, we processed 10- by 10-m resolution Landsat images of the Gil Gonzalez watershed with the vegetation index TASSELED CAP (Crist and Ciccone, 1984). TASSELED CAP values represent a gradient of forest and tree density ranging from pasture to dense, multiple-canopy forests. Because this study was done in conjunction with a riparian vegetation study, TASSELED CAP analysis and subsequent plots were within 100 m of perennial or ephemeral streams. The TASSELED CAP values were separated into five ordinal bins with an equal number of plots. Areas with 30 by 30 m of the same bin value were deemed potential plots. Eleven plots in each bin were randomly chosen and then screened based on the following criteria: (i) accessible within 1 km of a road, (ii) located outside an active river channel, and (iii) on a flat or moderate plot slope (slope $<10^\circ$) for hydrologic measurements. We selected three plots from each bin to stratify soil texture and livestock impact. To establish soil texture stratification in each plot, we performed field texture-by-feel tests and chose plots with a range of texture in each bin. While a gradient of soil textures exists in each bin, due to forests dominating riparian areas lower in the watershed with coarser soils, Bins 4 and 5 were the only bins with coarse soils (sand $>30\%$). To stratify livestock impact in each bin, we chose at least one plot with no or minimal livestock impact and one plot with higher intensity livestock impact. Lower bins generally included less woody vegetation and higher livestock impact, whereas higher bins included more woody vegetation and lower livestock impact (see LAI and hoofprint density in Table 1). Consequently, Bins 1 and 2 included agriculture, pasture, and savannah-like silvopastoral systems with sparse woody vegetation and no ungrazed areas, while Bins 3, 4, and 5 included grazed silvopastoral systems with higher woody vegetation density to ungrazed forests.

Data Collection

We estimated K_s using a Guelph permeameter (GP) (Soilmoisture Equipment Corp.). To determine the appropriate number of GP measurements per plot, we performed a power analysis on K_s data from the first three plots. Within each randomly selected 30- by 30-m plot, we established two transects separated by 10 m. Along each transect, we measured K_s 2 m laterally on either side every 4 m. We took 27 measurements in Plot 1 and 28 measurements in Plots 5 and 14. The power analysis established that 25 measurements were sufficient to explain the variability of K_s in each plot. In the remaining 12 plots, we took 25 equally spaced (5 m) measurements on a square grid in the 30- by 30-m plot. For each measurement and estimation of K_s , we followed standard field procedure for taking GP measurements at the soil surface (Reynolds, 1993; Soilmoisture Equipment Corp., 2008), taking necessary precautions to reduce compaction of the soil and smearing of the well wall (Reynolds, 1993). We took our measurements during the rainy season to reduce the effect on K_s of soil shrink-swell cycles typical of Vertisols (Messing and Jarvis, 1990). Measurements were not taken within 2 d of rainstorms with ≥ 2 cm precipitation to allow the soils to approach field capacity.

To quantify the woody vegetation structure, we measured the diameter at breast height (DBH) and LAI. The DBH of all trees in the plot with DBH >10 cm were summed for a total plot basal area (BA). We took two LAI measurements with a LAI-2000 plant canopy analyzer (Li-Cor, 1991) within each plot between the months of March and August 2010 during both the wet and dry seasons and averaged the values for the final plot LAI value. TASSELED CAP values in our study were highly correlated with LAI ($p < 0.0001$ for all three TASSELED CAP indices) and BA ($p < 0.0001$ for two TASSELED CAP indices and $p = 0.0004$ for the other); therefore, TASSELED CAP values accurately represented increased vegetation biomass, forest stand density, and forest cover (Chávez, 2011). The initial riparian forest study did not contain plots that fit the criteria for this study for Bin 1, so two pastures

Table 1. Untransformed data for each plot including the TASSELED CAP index (bin), plot description, soil physical properties including sand and clay contents, bulk density (BD), soil organic matter (SOM), and saturated hydraulic conductivity (K_s), livestock impact determined as hoofprint density, and vegetation variables leaf area index (LAI) and total plot basal area (BA) that were retained for single-variable stepwise regression (variables with VIF scores <10).

Plot	Bin	Description	Sand	Clay	BD	SOM	Hoofprint density	LAI	BA	K_s
			———%———		g/cm ³	%	prints/m ²	——m ² /m ² ——		log(mm/h)
1	1	pasture	6.5 (0.6)†	71.6 (2.9)	0.91 (0.03)	7.9 (0.56)	15.2	0.74	0	−0.73 (0.87)
2	1	pasture	12.5 (1.1)	56.5 (2.7)	1.27 (0.02)	8.0 (0.17)	5.1	0.74	0	−0.09 (0.60)
3	1	plantain field	14.1 (1.1)	51.8 (3.5)	1.09 (0.09)	8.6 (2.3)	0.2	4.60	0	0.75 (0.66)
4	2	pasture with small trees	17.0 (2.1)	59.9 (2.9)	1.20 (0.04)	6.0 (0.54)	22.7	1.81	0.00024	0.02 (0.77)
5	2	pasture with small trees	24.8 (6.5)	44.7 (10.6)	1.04 (0.08)	7.5 (0.93)	1.0	3.94	0.00035	0.91 (0.78)
6	2	corn field (with trees)	8.5 (0.9)	63.4 (2.4)	1.04 (0.04)	7.6 (0.69)	3.2	1.92	0.00025	0.11 (0.68)
7	3	forest-short and tall trees	18.7 (2.0)	41.8 (1.7)	1.18 (0.08)	7.0 (0.49)	2.1	3.13	0.0010	1.26 (0.40)
8	3	forest– short trees	16.0 (1.9)	62.8 (2.4)	1.12 (0.06)	8.0 (0.79)	12.9	2.90	0.00035	−0.12 (0.98)
9	3	forest-short and tall trees	10.3 (1.8)	54.9 (2.5)	1.20 (0.03)	6.8 (0.58)	0.6	3.52	0.0011	0.86 (0.42)
10	4	forest-short and tall trees	37.7 (3.8)	26.8 (1.6)	1.21 (0.04)	5.8 (0.44)	0.0	4.00	0.0016	0.93 (0.69)
11	4	forest-short and tall trees	20.2 (5.8)	55.6 (8.5)	0.87(0.04)	9.3 (0.57)	0.0	3.62	0.0023	0.91 (0.84)
12	4	forest-short and tall trees	33.2 (4.3)	41.5 (2.2)	1.17 (0.07)	5.7 (0.81)	4.2	4.24	0.0013	0.99 (0.68)
13	5	forest-short and tall trees	45.3 (15.1)	26.3 (6.8)	1.12 (0.06)	6.7 (1.2)	0.0	5.42	0.0024	1.35 (0.69)
14	5	forest-short and tall trees	15.4 (3.5)	58.1 (7.8)	0.90 (0.08)	11.0 (1.1)	0.03	5.54	0.0059	1.59 (0.77)
15	5	forest-short and tall trees	62.2 (6.0)	18.8 (2.1)	1.20 (0.04)	4.6 (0.66)	0.0	5.22	0.0031	0.97 (0.71)

† Plot-level standard deviations in parentheses except for plot-level variables (hoofprint density, LAI, and BA).

and one plantain plot were chosen for Bin 1 plots. Due to the lack of woody vegetation in these three plots, the plot BA was 0 cm². To estimate LAI for pasture plots (Plots 1 and 2), we used averaged literature values for pastures in tropical regions (White and Terry, 1979; Buschbacher, 1984; McWilliam et al., 1993; Meirelles et al., 2011; Kalácska et al., 2004). For the plantain field (Plot 3), we regressed data from three studies of LAI vs. plantain/banana density (Turner, 1972; Robinson and Nel, 1986; Cattani et al., 2007) and measured the plantain density in our plot to estimate LAI.

We collected two sets of soil samples at each plot, both extracted with a 100-cm³ bulk density ring (Blake and Hartge, 1986). The first set contained 18 soil samples from nine equally spaced (10 m) points on a square grid at two depths (0–5 and 5–15 cm) for bulk density (BD), porosity, and void ratio. We additionally collected 18 soil samples from six equally spaced locations on a rectangular grid (5- and 10-m spacing) at three depths (0–5, 5–15, and 15–30 cm) for sand, silt, and clay fractions, soil organic matter (SOM), and particle density. We determined soil texture by doing a particle size distribution with the pipette method (Gee and Bauder, 1986) and the SOM by the loss-on-ignition method (Nelson and Sommers, 1996).

To estimate the impact of land use on K_s in our plots, we counted livestock (cow and horse) hoofprints and dung along three 10-m-spaced transects in each plot. We counted hoofprints by placing four 1- by 1-m quadrats at the first set of soil sample locations

(the nine 10-m-spaced locations in a square grid) for a total of 36 quadrats per plot. Quadrats were placed either 1 or 2 m (chosen randomly) out from the soil sample location in four perpendicular directions. We analyzed whether there were any spatial correlations with hoofprints and K_s estimates using a simple regression and found no statistical correlation; therefore, we summed the hoofprint data across the entire plot for the rest of the statistical analyses. We counted dung piles inside the plot to get a plotwide dung density value. The hoofprint and dung density values are referred to as hoofprint and dung henceforth.

Pedotransfer Function and Leaf Area Index

The Rosetta PTF (Schaap et al., 2001) has been widely used to translate soil properties into hydraulic parameters based on neural network analyses that maximize the use of input data (BD, sand, silt, and clay) to estimate soil hydraulic parameters such as K_s . We used sand, silt, clay, and BD data at each K_s measurement point to generate predicted K_s values with the Rosetta PTF. To see if Rosetta could better predict K_s by including LAI, we compared predicted K_s values from Rosetta with our observed K_s values across a gradient of LAI.

Preferential Flow Paths

To characterize preferential flow paths, we performed dye tracer studies in a pasture plot with no woody vegetation (Plot 1) and a forest plot (Plot 14). We used brilliant blue dye to observe preferential flow paths (Flury and Flühler, 1995). Both plots had high

clay contents (72 and 58%, respectively) and were within 0.5 km of each other. We allowed a 100-L mix of 1.25 g/L dye to drain into a 1-by 1-m area. We excavated the plots to a 1-m depth, taking cross-sections every 10 cm. All dye that reached below the bulk wetting front (approximately 20 cm below the surface for each plot) was documented as a preferential flow path with their total straight-line distance from the surface. We documented the cause of the appearance as invertebrate holes (i.e., faunal turbation), fine root (diameter <1 mm), small root (diameter = 1 mm–1 cm), medium root (diameter = 1–3 cm), large root (diameter >3 cm), soil structure characteristic (seam, crack, etc.), or unidentifiable.

Data Analysis

Both K_S and BA were logarithmically transformed for statistical analysis. Variables bound by 0 to 1 (clay, sand, silt, SOM, and porosity) were transformed as $X' = \arcsin[\sqrt{(0.01X)}]$ (Crawley, 2007). Dung and hoofprint were transformed by taking their square root (Crawley, 2007). We took soil samples in a consistent gridded fashion: six soil sample points for BD, porosity, and void and nine soil sample points for sand, silt, clay, and SOM. While each of these soil sample points was at a K_S point, to have soil data for each of the 25 K_S measurements, we interpolated the sand, silt, clay, SOM, BD, void ratio, and porosity data at the plot level in ArcGIS 9.3 using inverse-distance weighting. We tested whether the soil data at different depths or combinations of depths better predicted K_S with an Akaike information criterion (AIC) (Burnham and Anderson, 2004), but a depth-weighted average of all three soil sample depths (0–5, 5–15, and 15–30 cm) had the lowest AIC scores.

For the regression analyses, we used residual plots and normal Q - Q plots to identify and remove outliers and leverage points for statistical analysis (Crawley, 2007), which resulted in removing seven from the initial 383 K_S measurements. We then performed initial single regressions for each variable to compare the relative effects of woody vegetation, soil properties, and livestock on K_S . To further compare the complex interactions among these three factors and their effects on K_S , we performed a multiple regression analysis. We first eliminated collinear variables with variance inflation factor (VIF) scores of 10 or higher (Ott and Longnecker, 2010), which removed sand, silt, void ratio, and porosity. Next, to visualize complex interactions among the variables and their strength of relationship with K_S , we used a nonparametric regression tree model to generate a tree diagram to illustrate complex interactions among the remaining variables (Breiman et al., 1984). We performed two multiple

stepwise regressions: (i) all remaining variables (LAI, BA, clay, BD, SOM, hoofprint, and dung), and (ii) the three variables in each variable category (woody vegetation, soil, and livestock) with the highest single regression R^2 value (LAI, clay, and hoofprint) and interaction terms between each of those three variables. Both procedures removed the terms with the highest p value until only nonsignificant terms remained. We performed a bootstrap analysis with the first stepwise regression to verify that the large sample size ($n = 376$) did not influence the significance of the model. Finally, to visualize the effect of woody vegetation on K_S across gradients in soil properties and livestock impact, we generated sets of three single regressions of K_S vs. LAI separated out into tertiles (thirds) of clay and hoofprint.

Results

Factors Affecting Saturated Hydraulic Conductivity

Leaf area index had the highest single regression correlation with K_S of all variables at an $R^2 = 0.331$ (Fig. 2A; Table 2). Basal area had a slightly lower R^2 value of 0.224, yet LAI and BA are known to be highly cross-correlated. Soil texture measurements of clay and sand had lower R^2 values than LAI, at 0.291 and 0.214, respectively. Both hoofprint density and dung were correlated with K_S , with R^2 of 0.291 and 0.190, respectively. Interestingly, K_S showed a nonsignificant trend with BD ($R^2 = 0.005$). Saturated hydraulic conductivity also correlated with increasing TASSELED CAP bin number (Fig. 2B).

The importance of LAI in predicting K_S is further emphasized by the nonparametric regression tree diagram (Fig. 3). An LAI of 3.015 separated high and low K_S values, and the top branches show that LAI separates most of the variability observed in the K_S data. Clay and sand separate the regression tree in both the high and

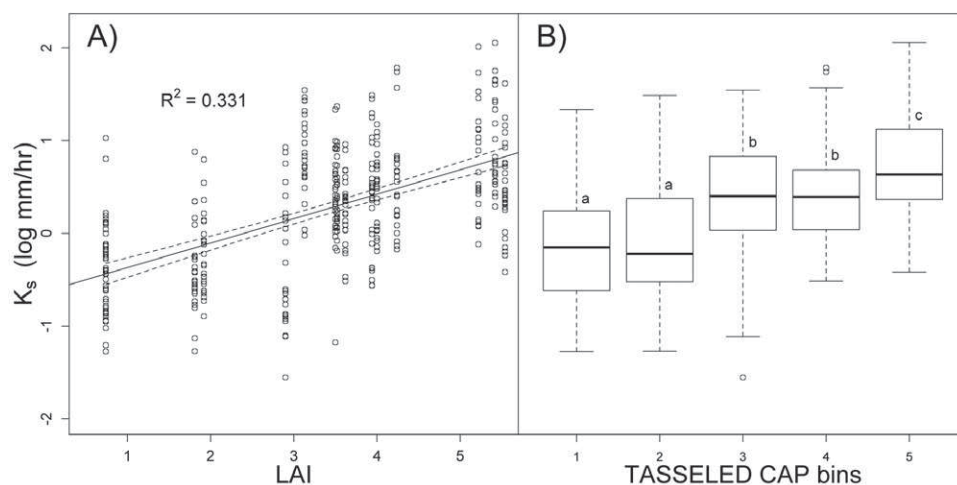


Fig. 2. Single regression of (A) saturated hydraulic conductivity (K_S) vs. leaf area index (LAI) and (B) K_S vs. TASSELED CAP bins. Significant differences ($p < 0.05$) between bins indicated by letters.

Table 2. Parameter estimates for predicting the saturated hydraulic conductivity (K_S) and p and R^2 values for each single regression ($n = 376$).

Variable	K_S estimate	R^2	p
Leaf area index	0.27	0.331	<0.001
Basal area	0.58	0.224	<0.001
Sand content	1.81	0.214	<0.001
Silt content	1.90	0.041	<0.001
Clay content	-2.32	0.291	<0.001
Soil organic matter	-2.57	0.015	0.016
Bulk density	0.40	0.005	0.164
Porosity	-27.7	0.014	0.021
Void ratio	-0.23	0.0076	0.091
Hoofprint density	-0.25	0.291	<0.001
Dung	-0.17	0.190	<0.001

low LAI categories, with low clay and high sand being associated with high K_S values and high clay and low sand with low K_S values.

The initial stepwise regression included all seven of the single variable terms with a VIF <10 (LAI, BA, clay, BD, SOM, hoofprint, and dung). The model with the lowest AIC (-456.6) had an R^2 of 0.395 and included the LAI, clay, and hoofprint terms, all with individual p values of <0.001 (Table 3). The next three models with the lowest AIC scores all had four terms and included LAI, clay, and hoofprint in addition to BD, SOM, or dung. The single-term model with the lowest AIC was LAI, with a score of -420.6, underscoring the prominence of LAI in predicting K_S . The bootstrap method on the lowest overall AIC model (clay, LAI, and hoofprint) revealed only a slight deviation from the original R^2 value—from 0.395 to 0.383. This means that despite the high number of observations ($n = 376$), this multiple regression model is robust (Crawley, 2007).

For the other multiple stepwise regressions, we used the LAI, clay, and hoofprint variables because they each had the highest R^2 value

Table 3. Results of stepwise regressions with the single-term multiple regression with the lowest Akaike information criterion (AIC) and single (leaf area index [LAI], clay content, and hoofprint density only) and interaction terms. Parameter estimates for predicting saturated hydraulic conductivity (K_S) and p values from stepwise regression with each variable.

Model	Final terms	Slope	p
Single terms	(intercept)	0.83	<0.001
	clay	-1.09	<0.001
	hoofprint	-0.088	0.002
	LAI	0.12	<0.001
Single and interaction terms	(intercept)	0.35	0.14
	clay	-0.74	0.002
	LAI	0.17	<0.001
	LAI $\times$ hoofprint	0.26	<0.001
	clay $\times$ LAI $\times$ hoofprint	-0.33	<0.001

for their variable category (Table 2). To decipher interactions among woody vegetation, soil properties, and livestock impact, the second stepwise regression included LAI, clay, and hoofprint as well as the two- and three-way interaction terms:

$$K_S = \text{LAI} + \text{clay} + \text{hoofprint} + (\text{LAI} \times \text{clay}) + (\text{clay} \times \text{hoofprint}) + (\text{LAI} \times \text{hoofprint}) + (\text{LAI} \times \text{clay} \times \text{hoofprint})$$

The model after removing nonsignificant terms in stepwise fashion was

$$K_S = 0.17\text{LAI}^{***} - 0.74\text{clay}^{**} + 0.26(\text{LAI} \times \text{hoofprint})^{***} - 0.33(\text{LAI} \times \text{clay} \times \text{hoofprint})^{**} + 0.35$$

where *, **, and *** are used to show significance at the $\alpha = 0.05$, 0.01, and 0.001 levels, respectively. This final model had an R^2 of 0.422. In both stepwise regression models, single clay and LAI terms remained, with LAI having a lower p value (Table 3). Even though hoofprint as a single variable has a negative slope with K_S (Table 3), the LAI $\times$ hoofprint interaction remained in the model and had a positive slope (Table 3), indicating that LAI increases K_S despite the negative trend of hoofprint. The three-way interaction also remained, with a negative slope indicating a complex interaction among the three primary variables.

The multiple stepwise regressions revealed complex interactions among LAI, clay, and hoofprint. To further elucidate how woody vegetation positively affects K_S , we compared regressions between LAI and K_S with clay and hoofprint broken out into thirds (tertiles): clay = 15 to 43, 43 to 59, 59 to 75%; hoofprint = 0 to 0.2, 0.2 to 1.8, 1.8 to 4.8 (Fig. 4). The slope of the middle and last tertile were significant (both $p < 0.001$), but the first tertile was not significant ($p = 0.12$). The regression line of LAI onto K_S increased with increasing clay content (Fig. 4), and the slope confidence intervals did not overlap between the 43 to 59 and

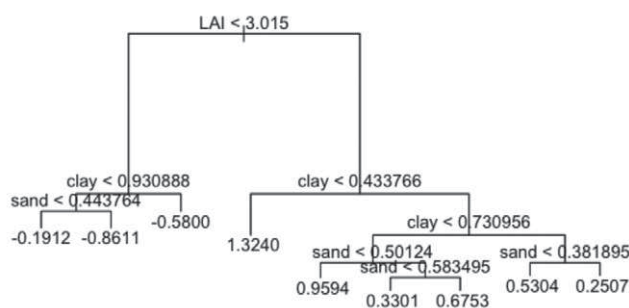


Fig. 3. A nonparametric regression tree model of the explanatory variables after variables with variance inflation factor (VIF) >10 are removed. Leaf area index (LAI) is clearly a dominant factor correlated with saturated hydraulic conductivity (K_S). High and low LAI values (split at 3.015) separate K_S into two clear bins.

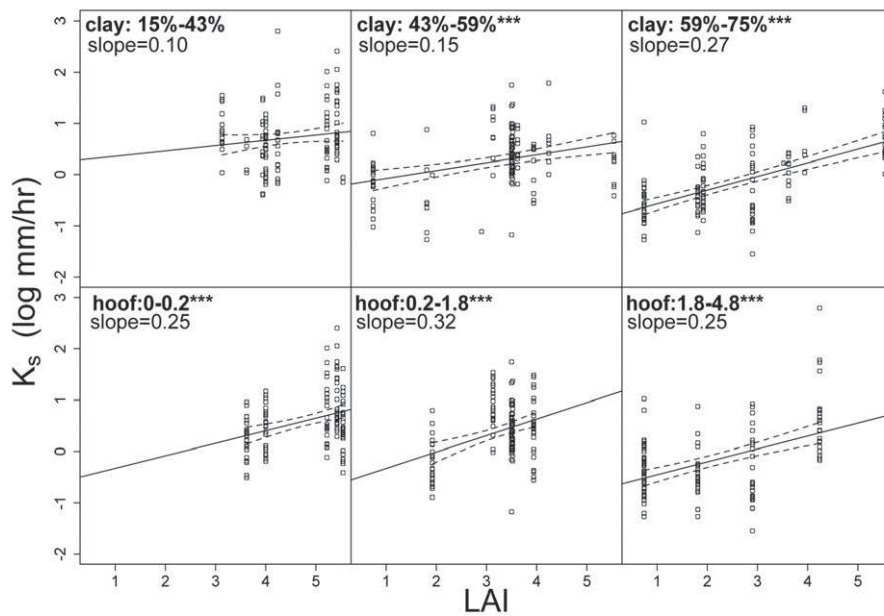


Fig. 4. Regressions of saturated hydraulic conductivity (K_s) and leaf area index (LAI), with data divided up into tertiles (thirds) of clay (top) and hoofprint density (bottom). ***Significant at $p < 0.001$. Slope estimates for each regression are included.

59 to 75% tertiles but did overlap between the 15 to 43 and 43 to 59% tertiles, signifying that the 59 to 75% tertile slope was significantly different than the slopes of the other two tertiles. This shows that the positive relationship between LAI and K_s varies with clay content. Conversely, the slopes of the K_s vs. LAI regression line in the hoofprint tertiles changed minimally across the different tertiles (Fig. 4), and the slope confidence intervals overlapped for all three tertiles.

with sand, silt, clay, and BD being accounted for, LAI tends to increase K_s .

Preferential Flow Path Analysis

The dye study revealed a clear difference in preferential flow paths in the forest compared with the pasture (Table 4). In the pasture, we found six dyed preferential flow paths below the initial wetting front dye line, while in the forest we found 51 dyed preferential flow paths past the wetting front (Fig. 6). The pasture plot contained three preferential flow paths due to soil structure characteristics,

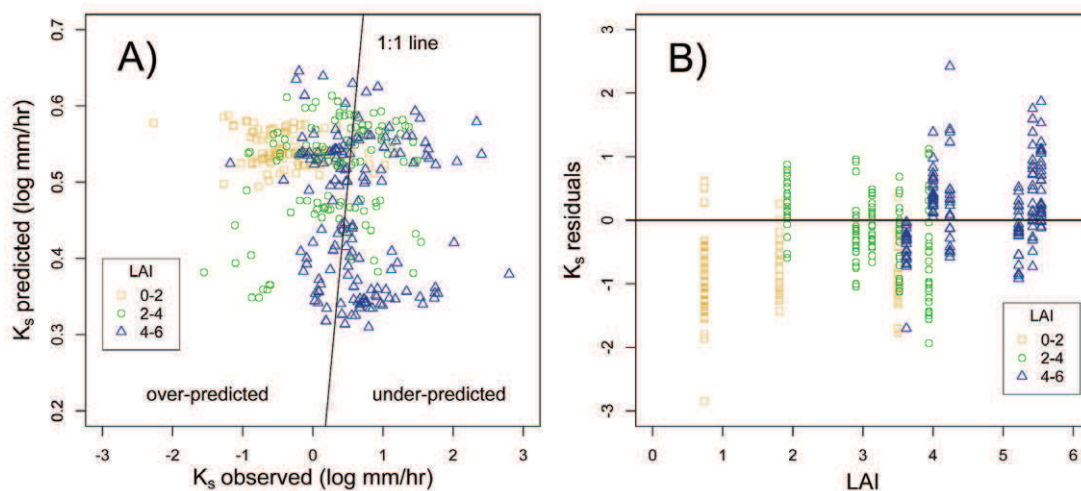


Fig. 5. (A) Predicted saturated hydraulic conductivity (K_s) from Rosetta pedotransfer function vs. observed K_s and from our study, with the leaf area index (LAI) of each data point signified by color and shape (yellow squares: LAI < 2; green circles: $2 < \text{LAI} < 4$; blue triangles: LAI > 4). Points where Rosetta overpredicted the K_s value are above the 1:1 line and where it underpredicted they fall below the 1:1 line; and (B) residuals (residuals = observed K_s - predicted K_s) plotted against LAI, with negative residuals below the 1:1 line and positive residuals above the 1:1 line.

Pedotransfer Function and Leaf Area Index

We graphed the predicted K_s values from the Rosetta PTF against the observed K_s , with color and shape coding to indicate low (yellow square: LAI < 2), medium (green circle: $2 < \text{LAI} < 4$), or high (blue triangle: LAI > 4) based on LAI values (Fig. 5A). The K_s measurements in plots with low and medium LAI predominantly overpredicted K_s 75 and 80% of the time, respectively. Conversely, K_s measurements in plots with high LAI values were only overpredicted 32% of the time, meaning that the majority were underpredicted. The residuals (observed K_s - predicted K_s), graphed in Fig. 5B, showed a general negative bias for K_s measurements in low-LAI plots and a positive bias for K_s measurements in high-LAI plots. The higher percentage of positive residuals for plots with high LAI values than for low LAI values suggests that

Table 4. Number of preferential flow paths evidenced by dye presence past the dyed wetting front and quantities counted of each preferential flow path.

Cause of dye appearance	Flow path count	
	Pasture plot (Plot 1)	Forest plot (Plot 14)
	no. —————	
Soil structure	3	15
Invertebrate hole	2	1
Fine (<1-mm) root	0	10
Small (1 mm–1 cm) root	0	11
Medium (1–3-cm) root	1	9
Large (>3-cm) root	0	3
Unidentifiable	0	2
Total	6	51

two from invertebrate holes, and one from a medium-sized root. The forest plot had 15 preferential flow paths from soil structure characteristics and one insect hole, but the majority of the preferential flow paths were from roots: 10 from fine roots, 11 from small roots, nine from medium-size roots, and three from large roots.

Discussion

Our study illustrates that the density of standing woody vegetation significantly increases K_S (Fig. 2 and 5), thereby increasing the ability of the soil to infiltrate water and improve the hydrologic functions of the soil at our sites. We also observed that the effect of woody vegetation on K_S is greater in fine-textured soils than in coarse-textured soils. The regression slopes of LAI and K_S increased for the two finer soil tertiles (Fig. 4), and the regression slopes for those two tertiles were significant. The coarse-soil tertile was not significant. While we could not find low LAI, high sand plots, however, our results still suggest that LAI has little or no effect on K_S in coarse soils but does have an increasing effect on finer soils. While coarse soils in general show a higher K_S than fine-textured soils (Saxton et al., 1986; West et al., 2008; Senarath et al., 2010), our results suggest that woody vegetation is a mechanism in increasing K_S in fine-textured soils where root-macropore formation significantly increases K_S compared with what would be expected from the soil texture class alone.

The robustness of woody vegetation in maintaining high K_S across livestock impact is less clear. Some evidence supports woody vegetation maintaining a high K_S despite livestock impact. In the multiple regression, the two-way LAI $\times$ hoofprint term was significant with a positive slope, indicating that when LAI and hoofprint density are both increasing, there is still an increasing relationship with K_S . Additionally, some plots (7 and 12) had higher LAI, moderate grazing, and still had a high K_S . Giertz

and Dieckkrüger (2003) similarly found that soils with woody vegetation can maintain a relatively high K_S despite moderate agricultural use. Contrarily, in the multiple regression the three-way interaction term clay $\times$ LAI $\times$ hoofprint was significant with a negative slope, indicating that with high clay and hoofprint density, there is a trend to decreasing K_S despite increasing LAI. During heavy rain events where ponding and overland flow occurred in pastures, water ponding was also observed in heavily used livestock paths in dense forests (R. Niemeyer, personal observation), suggesting a threshold of livestock impact where high LAI forest plots do not maintain a high K_S . Still, it appears that livestock impact on the surface K_S may be minimized if woody vegetation remains on the landscape.

There was a correlation with K_S and increasing clay and decreasing sand, as has been commonly observed (Saxton et al., 1986; West et al., 2008; Senarath et al., 2010). However, we found no significant relationship between bulk density and K_S (Table 2). Other hydrologic studies in the tropics have showed a positive correlation between BD and reduced K_S and infiltration (Schack-Kirchner et al., 2007; Mehta et al., 2008). This could be due to the unique nature of Vertisols and how land use affects their structure.

Comparing our findings with those presented by Thompson et al. (2010) suggest some disagreement with the difference in the factors that drive differences in K_S . Thompson et al. (2010) concluded that biomass increases infiltration capacity in xeric sites but not in mesic or hydric sites. Because our study was on a xeric site according to their classification (pan evaporation/precipitation > 1), our study confirms their results that biomass will increase the infiltration capacity in xeric sites; however, our study also reveals a more nuanced relationship between K_S –infiltration capacity, woody vegetation–biomass, and soil texture. In their study, the xeric sites were mostly fine soils: of the 92 xeric sites with soil data in their study, 43% had

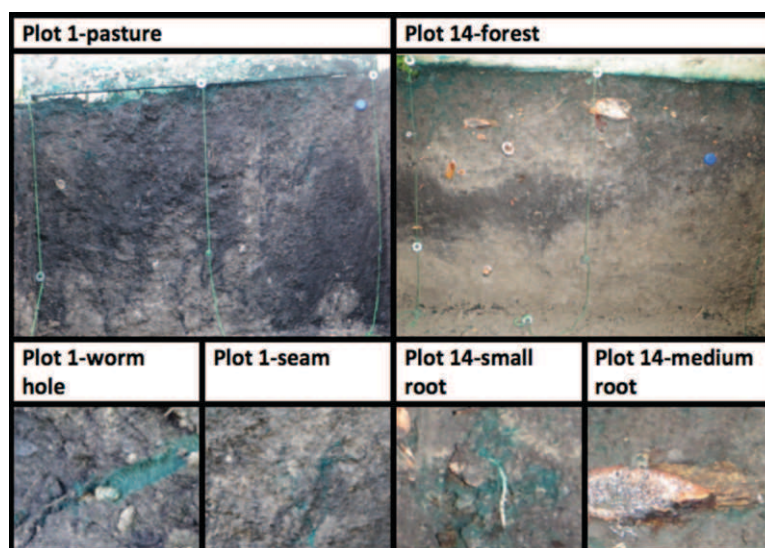


Fig. 6. Soil profiles after dye infiltration. Plots were with 0.5 km of each other. The cause of dye presence (e.g., worm pore, seam, etc.) is identified.

a high clay content (clay content >40%), and only 24% had a low clay content (clay content <20%). Contrarily, of the 105 mesic and hydric sites with soil data, 5% had high clay content and 71% had low clay content. They saw no effect of biomass on infiltration in the coarse-soil mesic and hydric sites and an increasing effect of woody vegetation on the finer xeric sites. Our study showed that woody vegetation will increase K_S in fine soils but to a lesser extent in coarse soils (Fig. 4), so we would expect the finer soils of xeric sites to show a stronger relationship between biomass and infiltration capacity than the coarser soils of the mesic and hydric sites. We suggest that they might not have found a strong relationship between biomass and infiltration capacity in mesic and hydric sites because of the interaction between soil texture and woody vegetation. Other studies have confirmed that for fine soils in mesic and hydric sites, woody vegetation will be a greater determinant of K_S than soil texture. Hassler et al. (2011), in a tropical rainforest (mesic–hydric), saw similar K_S values across a range of soil types—meaning that woody vegetation was the main driver of K_S , not soil texture. Similarly, Sobieraj et al. (2002) in a tropical rainforest (mesic–hydric) did not find a correlation with K_S and soil type, and attributed this lack of correlation with soil type to biotic causes increasing the macroporosity and thereby increasing K_S . Both of these studies approximated K_S across a range of soil types that were all fine (clay >25%). They confirmed that in finer textured soils, woody vegetation can be a greater driver of K_S than soil texture.

Further support that woody vegetation does have an effect on K_S is seen in our PTF analysis (Fig. 5B). Our results indicate that including woody vegetation density or other vegetation metrics (e.g., LAI, biomass, etc.) in PTFs to predict K_S would improve their predictive power. Currently, the Rosetta PTF does not explicitly include woody vegetation or other biotic parameters to predict K_S . This finding is further supported by Spaeth et al. (1996), who found that the inclusion of vegetation significantly improved predictions of infiltration capacity compared with predictions solely based on soil physical properties. It is important to bring these findings into use in PTFs, which are increasingly used in research and management to predict how deforestation, agriculture, and climate change will alter hydrologic processes. While more data are needed to determine how woody vegetation structure affects soil hydrologic properties in the tropics, future PTFs could incorporate some measure of woody vegetation with physical soil properties as determinants of K_S in these models.

The dye study suggests that preferential flow paths are a probable mechanism for how woody vegetation and forest structure increases K_S . Bioturbation and the resultant macroporosity from roots compared with fauna appear to be a greater source of preferential flow paths in the forest site; however, a more extensive study is needed to confirm these results. Even though our study was limited to two plots, other studies confirm these results. Mbagwu (1997) found that in Nigerian dry tropical soils of micro- (<10 μm), meso- (10–1000 μm), and preferential (>1000

μm) porosities, preferential porosity had the highest correlation coefficient with K_S , followed by mesoporosity. These findings were confirmed by our dye study, where roots larger than fine were the source of most of the preferential flow paths in the forest that had a higher K_S . This suggests that meso- and preferential porosities due to roots are primary infiltration pathways in conducting water in the soil, thereby increasing the overall K_S of the soil regardless of soil texture. Bonell et al. (2010) saw that areas that had previously been degraded for decades to centuries and then converted to tree plantations had K_S values that were closer to the background K_S (the K_S of local forests) at the surface but not at deeper points in the soil profile. They suggested that vegetation processes mediate the surface recovery of background K_S . Our dye study partially confirmed that indeed biological processes increase preferential flow paths occurring in the forest compared with the pasture and could mediate a recovery of degraded landscapes to background K_S .

Conclusion

Many tropical areas face a twofold pressure of scarce financial resources and a need to maintain or improve water resources in complex landscapes where mismanagement has caused increased flooding, reduced natural water filtration capacity, and decreased base flows (Bruijnzeel, 2004). Our study illustrates that woody vegetation increases K_S and thereby the ability of the soils to facilitate infiltration and reduce runoff in dry, tropical Nicaragua. The correlation between woody vegetation (LAI) and K_S was particularly present in finer textured soils and was robust to moderate livestock impact but not to heavy livestock impact. A forested site had eight times more preferential flow paths than a pasture site, and roots were the primary source of preferential flow paths in the forest site, suggesting a mechanism for increased K_S in sites with woody vegetation. This information is helpful for managers to predict where woody vegetation maintenance and restoration could likely have the largest impact on rehabilitating soil hydrologic functions. The incorporation of woody vegetation to predict K_S with the Rosetta PTF showed that K_S prediction would be improved by including woody vegetation.

In our study, LAI served as a good predictor of areas with high and low K_S based on vegetation metrics and therefore a good predictor of which areas are more or less vulnerable to surface runoff and erosion (e.g., Giertz and Dieckkrüger, 2003). Remote sensing (e.g., TASSELED CAP) could be an inexpensive method to predict areas of high and low LAI and therefore areas likely to have high or low K_S . This information could be used to target high-priority areas for management efforts to increase or maintain woody vegetation and thereby reduce surface runoff and erosion while protecting water resources in tropical areas (Fremier et al., 2013).

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